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Comparing the effectiveness of LLM-Powered coaching with human coaching and GPT conversation

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ABSTRACT

This study examines the effectiveness of LLM-powered coaching compared to traditional human coaching and GPT-4 interactions across 11 key criteria of coaching effectiveness. Leveraging a randomized controlled trial with 32 participants, the research revealed LLM-powered coaching significantly outperforms GPT-4 in empathy, engagement, and encouragement – achieving outcomes comparable to human coaching in all dimensions. Notably, LLM-powered coaching showed the greatest improvements in Positive Affect (+8.49%) and reductions in Negative Affect (–27.45%) among the groups. Qualitative feedback highlights the strengths of human coaching in emotional depth and nonverbal cues, while AI coaching excels in its accessibility, flexibility, and structured support for self-reflection and action planning. This research underscores the transformative potential of AI-powered coaching to foster individuals' growth and well-being on a large scale. By offering accessible, scalable, cost-effective coaching solutions, AI coaching complements traditional human coaching, expanding access to personal and professional development for a broader and more diverse population.

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Coaching; coaching psychology; large language models; coaching effectiveness; coaching interventions

Introduction

Coaching has become a widely recognized tool for facilitating personal growth, enhancing performance, and improving well-being in various domains, from professional development to mental health. Research consistently demonstrates its positive impact on self-confidence, emotional resilience, and goal achievement (Grant, 2014; Theeboom et al., 2014). However, its accessibility is often limited by cost, availability, and scalability. Recent advancements in artificial intelligence (AI), particularly large language models (LLMs) like GPT-4, present a transformative opportunity to bridge these gaps.

Despite their potential, questions remain regarding their effectiveness compared to human coaches and their ability to address complex emotional and interpersonal challenges. This study sought to evaluate the effectiveness of an LLM-powered coach in comparison to traditional human coaching and GPT-4 interactions, with a focus on 11 key coaching dimensions such as engagement, empathy, psychological safety, and emotional well-being. Through a randomized controlled trial, this study explored both the strengths and limitations of LLM-powered coaching, contributing to the growing body of knowledge on AI's role in coaching and related fields.

Literature review

Coaching has long been recognized as a powerful tool for fostering personal and professional growth. As traditional coaching methods continue to demonstrate their effectiveness, technological advancements such as large language models (LLMs) present an opportunity to reimagine coaching delivery at scale. This literature review explores the fundamental principles and criteria of effective coaching while examining the emerging role of LLMs in coaching and related fields.

Coaching

Coaching is a process aimed at facilitating individuals' learning, growth, performance, and overall well-being (Rogers, 2016). It is widely recognized in both workplace and personal development contexts for its proven effectiveness in enhancing self-confidence, interpersonal skills, and performance (Moen & Federici, 2012; Mukherjee, 2012). Research consistently demonstrates its benefits: for instance, a meta-analysis study revealed significant positive impacts of coaching on employee performance, well-being, coping mechanisms, and goal-directed self-regulation (Theeboom et al., 2014). Similarly, Grant (2014) showed that coaching improves solution-

Table 1. Coaching evaluation criteria.

Dimension	Coaching Criteria	Description
Affect	Emotional Well-being	Ability to perceive, positively effect on, and moderate clients' emotions (Grant, 2010; Theeboom et al., 2014).
	Engagement	The client's emotional involvement and active participation in the coaching process, often resulting in a state of "flow" (Rogers, 2016).
	Empathy	Ability to understand, resonate with and respond to clients' emotions and perspectives (Niedtfeld, 2017; Passmore, 2010).
	Psychological Safety	Fostering a secure, supportive coaching environment that encourages risk-taking and learning through authentic listening, acceptance, and respect (Ostrowski & Potter, 2023; Rogers, 2016).
	Satisfaction	Client's overall evaluation of the coaching experience, including the quality of the coaching process, the relationship with the coach, and the outcomes achieved (Boyce et al., 2010; O'Broin & Palmer, 2018; Whybrow & Nottingham, 2018).
	Encouragement	Encouraging and supporting clients' growth (Passmore, 2010).
Cognition	Self-Awareness	Facilitating clients' exploration, reflection, and evaluation of their knowledge, skills, emotions, strengths, behaviors, and cognitive patterns (Caruso et al., 2015; Church, 1997; Passmore, 2010; Ting & Hart, 2004).
	Insightfulness	Inspiring clients to adopt alternative perspectives and develop strategies for growth through reflective thinking (Grant, 2018).
	Problem-Solving	Fostering clients' problem-solving skills, enabling them to tackle obstacles independently and become more resourceful and adaptive over time (Passmore, 2010).
Behavior	Goal Setting	Facilitating clients in setting meaningful goals enables focused action and drives significant behavioral change (Rogers, 2016).
	Action Planning	Creating structured task strategies to guide, motivate, and keep clients on track toward meaningful goals (Grant, 2018).

focused thinking, goal attainment, self-efficacy, and resilience while also reducing symptoms of depression.

Effective coaching is characterized by adherence to specific criteria. According to Passmore and Theeboom (2016), coaching psychology focuses on understanding and facilitating behavioral, cognitive, and emotional changes that occur within the coaching process. In addition, Ely et al. (2010) outlined five key evaluation criteria for coaching – reactions, cognitive learning, affective learning, behavior, and results. In a broader meta-analysis, Jones et al. (2016) highlighted coaching's positive effects across four key outcomes: affective, cognitive, skill-based, and individual-level results.

Drawing from prior research, this study will highlight 11 critical coaching criteria categorized into affective, cognitive, and behavioral dimensions (see Table 1). These include emotional well-being, engagement, empathy, psychological safety, satisfaction, encouragement, self-awareness, insightfulness, problem-solving, goal setting, and action planning. These 11 criteria of coaching serve as a comprehensive framework to assess the effectiveness of coaching interventions, providing a structured lens through which to evaluate the transformative potential of different coaching modalities. This study employs these criteria to compare the efficacy of LLM-powered coaching with traditional human coaching and GPT-4 interactions, illuminating the unique strengths and challenges of each approach. The next section delves into the emerging role of large language models (LLMs) in coaching and related fields, exploring their capabilities and transformative potential.

Large language models (LLMs) in coaching

Large Language Models (LLMs) represent one of the most significant advancements in Artificial Intelligence (AI) in recent years, showcasing a broad spectrum of capabilities applicable to coaching and related fields. Those models are based on Transformer architecture, a process that involves training an AI on large web-text data with a self-supervised objective of predicting the next word in a partial sentence (Bubeck et al., 2023). LLMs like GPT-4 have demonstrated the capacity to reason, plan, and adapt based on experience, excelling in complex tasks across domains such as mathematics, law, medicine, and psychology (Bubeck et al., 2023).

Notably, LLMs have shown promise in psychology. For instance, Kosinski (2023) demonstrated that GPT-4 could successfully complete 95% of Theory of Mind (ToM) tasks, reflecting an emergent ability to understand others' thoughts and feelings – an essential component of social competence and interaction (Carlson et al., 2013; Leslie et al., 2004). Additionally, Blyler and Seligman (2023a, 2023b) revealed that LLMs can support self-discovery and foster personal growth and well-being. In a recent study, Xie and Ostrowski (*in press*) highlighted that more than half of the participants in their study expressed a strong willingness to utilize LLM-powered coaching for professional challenges, mental well-being, academic stress, and relationship issues. These findings underscore the transformative potential of LLMs in coaching.

While LLMs demonstrate significant promise, their effectiveness in coaching and related fields remains a topic of ongoing debate. Proponents argue that AI-powered chatbots can have a positive and transformative impact on coaching practices. For example,

Terblanche, Moly, De Haan, and Nilsson (2022) suggest that AI coaching effectively supports specific outcomes such as goal attainment and holds the potential to democratize access by making coaching more affordable and widely available. In a recent study, Cho et al. (2023) demonstrated that LLMs can offer empathetic, adaptable, and engaging conversations, holding promise in psychological counseling. Another research indicated that AI-driven CBT interventions were as effective as traditional methods in facilitating therapeutic progress and even more effective at reducing user anxiety immediately following sessions (Kian et al., 2024). Moreover, Arakawa and Yakura (2024) demonstrated that an LLM-powered chatbot could facilitate deep self-reflection in executive coaching and suggested that blending the strengths of human and AI coaching provides an optimal approach to fostering reflective practices and personal growth.

In contrast, other researchers, such as Bachkirova and Kemp (2024), argue that many aspects of effective coaching are unique, abstract, and challenging to measure or replicate. While AI can simulate understanding by recognizing linguistic patterns, it lacks true empathy, moral reasoning, and personal experience, limiting its capacity to fully replicate human coaching dynamics (Bachkirova & Kemp, 2024). Chiu et al. (2024) further emphasize that current LLMs tend to display characteristics more common in low-quality therapy, such as providing problem-solving advice during emotional disclosures, which deviates from recommended practices. Although LLMs reflect on clients' strengths and needs, they often do so without the nuanced skill set that human coaches bring. Thus, systematic frameworks for assessing and enhancing the quality of LLM-based mental health support are urgently needed. To address this issue, this study offers an empirical evaluation of an LLM-powered AI coach, Hey Sherry, in the real world and compares its effectiveness with human coaching and GPT interactions.

Methods

This study evaluated the effectiveness of an LLM-powered coach, Hey Sherry, in comparison to traditional human coaching and ChatGPT-4 interactions.

Hypotheses

- (1) **H1:** LLM-powered coaching will demonstrate greater effectiveness than ChatGPT-4 interaction.
- (2) **H2:** LLM-powered coaching will demonstrate comparable effectiveness to traditional human coaching.

Participants

Participants were recruited through a multi-pronged approach, including email invitations distributed within the Organizational Dynamics Department at the University of Pennsylvania and advertisements on social media platforms such as LinkedIn, Facebook, Instagram, and WeChat. Institutional Review Board (IRB) approval was obtained from the University of Pennsylvania, ensuring that the study adhered to ethical research practices.

Participants were required to meet specific eligibility criteria. They had to be at least 18 years old, demonstrate an interest in personal development and emotional well-being, and have access to ChatGPT through either a computer or smartphone. However, individuals currently taking prescribed medication for a mental health condition were excluded from participation to ensure safety and avoid potential risks, as self-guided LLM interactions may not adequately address existing mental health conditions.

The study employed a rolling registration process to ensure enough participants were recruited. A total of 134 individuals registered for the study. Of these, seven were excluded due to ineligibility (medication use), two were unreachable after registration, and four withdrew before the study began. The remaining 121 participants were randomly assigned to one of three groups: LLM-powered coaching (Hey Sherry) ($n = 49$), human coaching ($n = 36$), or ChatGPT-4 interactions ($n = 36$). Out of these, 49 participants completed the pre-coaching surveys, and 32 participants completed the entire study. The final group distribution included 10 participants in the AI coaching group, 12 in the human coaching group, and 10 in the GPT-4 group.

The participants in the study represented a diverse demographic range, with ages spanning from 19 to 71 years. They also varied in terms of gender, educational background, and racial and ethnic identities. Statistical analyses revealed no significant differences (all p -values > 0.05) across the three study groups despite this demographic variation. Specifically, the analyses showed no significant differences in age ($F = 0.86$, $p = 0.42 > 0.05$), gender ($\chi^2 = 4.21$, $p = 0.38 > 0.05$), educational background ($\chi^2 = 2.26$, $p = 0.69$), or racial and ethnic composition ($\chi^2 = 3.1$, $p = 0.80 > 0.05$) (see Table 2). This demographic balance provided a robust foundation for making valid comparisons among the three groups.

Procedure

This study utilized a randomized controlled trial (RCT) design to compare the effectiveness of three coaching modalities: LLM-powered coaching (AI Coach), human

Table 2. Demographics of participants by groups.

Categories	AI	Human	GPT	Test Statistic	p-value
N	10	12	10		
Age (M(SD))	30.40(7.52)	38.67(18.23)	35.20(14.71)	F = 0.86	0.42
Gender				$\chi^2 = 4.21$	0.38
Female	70.00%	50.00%	50.00%		
Male	20.00%	50.00%	50.00%		
Non-binary/third gender	10.00%	0.00%	0.00%		
Educational Background				$\chi^2 = 2.26$	0.69
Master's Degree	70.00%	50.00%	60.00%		
Bachelor's Degree	30.00%	41.67%	40.00%		
Doctorate Degree	0.00%	8.33%	0.00%		
Race and Ethnicity				$\chi^2 = 3.1$	0.80
Asian	80.00%	66.67%	60.00%		
White	10.00%	16.67%	30.00%		
Black or African American	10.00%	8.33%	10.00%		
Hispanic, Latino, or Spanish origin	0.00%	8.33%	0.00%		

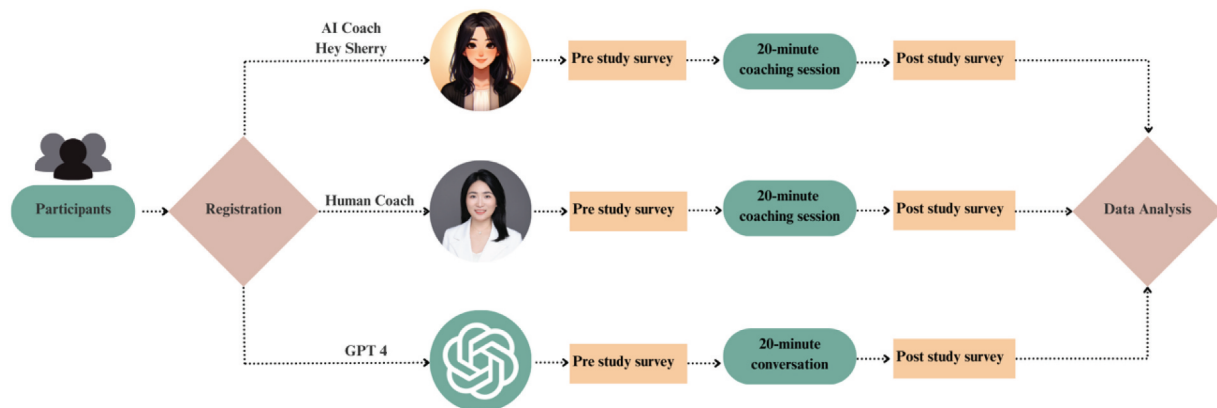
M = Mean; SD = Standard Deviation; F: one-way ANOVA result for age; χ^2 : Chi-square test results for gender, educational background, and race/ethnicity.

coaching, and GPT-4 interaction. Upon registering for the study, participants received emails with instructions to complete consent forms and pre-study surveys to collect demographics and baseline emotional measures. After completing the surveys, participants engaged in a 20-minute coaching intervention (see Figure 1 for study design flow):

- **AI coaching group:** Participants engaged in a 20-minute coaching session with Hey Sherry, a customized, prompt-tuned AI coach based on the GPT-4 model. Prompt-tuning involves optimizing the input prompts to guide the pre-trained model's behavior on specific tasks without changing the model's parameters (Demszky et al., 2023). The development of Hey Sherry involved three steps: establishing a coaching persona with prompts, creating a knowledge-based framework for AI coaching, and inputting a specific coaching dataset (Xie & Ostrowski, *in press*). This AI Coach is available 24/7 and supports more than 80 languages. To ensure privacy, all

coaching conversations were stored locally on participants' personal devices and were not shared with anyone else.

- **Human coaching group:** Participants engaged in a 20-minute human coaching session with a professional human coach. The coach held master's degrees in both Applied Positive Psychology and Organizational Dynamics from the University of Pennsylvania with a certificate in leadership and organizational coaching. The coaching sessions were held via Zoom, offered in both English and Chinese Mandarin, for an accessible, personalized, and inclusive coaching experience. To maintain confidentiality and privacy, none of the sessions were recorded.
- **ChatGPT group:** Participants engaged in a 20-minute conversation with a non-prompt-tuned large language model (LLM), GPT-4. All interactions were stored locally on participants' personal devices and were not shared with anyone else.

**Figure 1.** Study procedure.

After completing the coaching intervention, participants were instructed to complete post-study surveys. These surveys included an assessment of emotional well-being (Positive and Negative Affect), as well as a self-report evaluation of coaching effectiveness across 10 dimensions. This systematic procedure ensured a consistent and structured approach for assessing the comparative effectiveness of each coaching modality while minimizing variability due to differences in session format or duration.

Measures

To comprehensively evaluate the impact of coaching across different modalities, this study employed a combination of quantitative and qualitative measures. Emotional well-being was assessed using a validated psychological scale, while coaching effectiveness was evaluated through a self-report survey addressing multiple dimensions of the coaching experience. Additionally, open-ended questions provided deeper qualitative insights into participants' perceptions.

Emotional well-being

This study used the Positive and Negative Affect Schedule (PANAS) (Watson et al., 1988) to measure participants' emotional well-being changes before and after coaching conversations. The PANAS scale included 20 items, with 10 items measuring positive affect (interested, excited, strong, enthusiastic, proud, alert, inspired, determined, attentive, active) and 10 items measuring negative affect (distressed, upset, guilty, scared, hostile, irritable, ashamed, nervous, jittery, afraid). Each item is rated on a five-point Likert scale, ranging from 1 ('Very Slightly or Not at All') to 5 ('Extremely'), capturing the intensity of emotions experienced by participants (Watson et al., 1988). Both pre-and post-coaching assessments were conducted to examine changes in emotional states.

Self-report coaching evaluation criteria

The study employed a self-report survey (see Appendix A) to evaluate coaching effectiveness across the 10 criteria

detailed in the literature review above, encompassing three psychological dimensions: affect, cognition, and behavior. Participants evaluated these criteria using a five-point Likert scale (ranging from 1 = 'Strongly Disagree' to 5 = 'Strongly Agree'), with higher scores reflecting greater perceived effectiveness of the coaching session.

To complement the quantitative data, participants responded to three open-ended questions designed to capture qualitative feedback. These questions explored the perceived benefits of the coaching, areas for improvement, and additional reflections on the coaching experience. This mixed-methods approach allowed for a robust evaluation of coaching effectiveness, providing both numerical data and in-depth perspectives on the strengths and limitations of each coaching modality.

Results

An analysis of variance (ANOVA) was conducted to evaluate potential baseline differences among the three groups (AI Coaching, Human Coaching, and GPT 4 Conversation) on positive and negative affect measures prior to the coaching interventions. The results indicated no statistically significant differences across the groups for positive affect ($F(2, 29) = 0.25, p = 0.78$) or negative affect ($F(2, 29) = 2.09, p = 0.14$). These findings confirm that the groups were comparable in terms of their initial emotional states, ensuring that any observed post-coaching differences could be attributed to the intervention rather than pre-existing disparities.

Emotional outcomes

To evaluate the emotional impact of each coaching modality, paired t-tests were conducted within the AI Coaching, Human Coaching, and GPT groups. These analyses assessed changes in both Positive Affect (PA) and Negative Affect (NA) from pre- to post-coaching sessions (see Table 3).

Table 3. Positive affect and negative affect pre- and post-coaching.

Measure	Condition	Pre M(SD)	Post M(SD)	Change %	t	p-value	Effect Size d
Positive Affect	AI Coach	31.80 (6.25)	34.50 (7.92)	8.49%	-1.00	0.35	0.33
	Human Coach	31.83 (6.77)	32.42 (8.70)	1.83%	-0.22	0.83	0.07
	GPT	34.10 (11.82)	35.30 (11.90)	3.52%	-0.36	0.73	0.12
Negative Affect	AI Coach	20.40 (9.36)	14.80 (5.39)	-27.45%	1.48	0.17	-0.49
	Human Coach	14.75 (4.85)	13.50 (4.34)	-8.47%	0.92	0.38	-0.28
	GPT	19.40 (6.40)	17.10 (7.55)	-11.86%	1.08	0.31	-0.36

Pre=pre-test; Post=Post-test; M=Mean; SD=Standard Deviation; t= paired t-test results for pre- vs. post-coaching comparisons within each group; d= effect size: d = 0.2 (small), d = 0.5 (medium), and d = 0.8 (large) (Cohen, 1988).

Positive affect

Positive Affect was assessed before and after coaching intervention across the three groups to measure changes in participants' positive emotional states. The AI coaching group demonstrated the highest percentage increase in Positive Affect (8.49%), with post-coaching scores rising from a mean of 31.80 (SD = 6.25) to 34.50 (SD = 7.92). While the change was not statistically significant ($t = -1.00$, $p = 0.35$), the effect size was small ($d = 0.33$), indicating a moderate trend toward improvement. The GPT group exhibited a smaller increase in Positive Affect (3.52%), with post-coaching scores increasing from a mean of 34.10 (SD = 11.82) to 35.30 (SD = 11.90). The change was not significant ($t = -0.36$, $p = 0.73$), and the effect size was negligible ($d = 0.12$). The Human group showed the smallest improvement (1.83%) in Positive Affect, with scores increasing from 31.83 (SD = 6.77) to 32.42 (SD = 8.70). This change was not statistically significant ($t = -0.22$, $p = 0.83$), and the effect size was negligible ($d = 0.07$).

Negative affect

Negative Affect was measured to evaluate changes in negative emotional states pre- and post-coaching intervention. The AI group experienced the most significant percentage reduction in Negative Affect (-27.45%), with scores decreasing from a mean of 20.40 (SD = 9.36) to 14.80 (SD = 5.39). Although the change was not statistically significant ($t = 1.48$, $p = 0.17$), the medium

effect size ($d = -0.49$) suggests a meaningful trend toward improvement. The GPT group achieved a moderate reduction (-11.86%) in Negative Affect, with scores dropping from 19.40 (SD = 6.40) to 17.10 (SD = 7.55). This change was also not statistically significant ($t = 1.08$, $p = 0.31$), with a small effect size ($d = -0.36$). The Human group showed the smallest decrease in Negative Affect (-8.47%), with scores decreasing from 14.75 (SD = 4.85) to 13.50 (SD = 4.34). The change was not statistically significant ($t = 0.92$, $p = 0.38$), with a small effect size ($d = -0.28$). The visual representation in Figure 2 illustrates the table data, highlighting the AI group's larger reduction in Negative Affect compared to the Human and GPT groups.

Overall, the AI coaching group outperformed the Human and GPT groups in improving Positive Affect and reducing Negative Affect, as evidenced by higher percentage changes and larger effect sizes. While the observed trends are promising, the lack of statistical significance suggests that these findings should be interpreted cautiously, likely due to small sample sizes ($N = 32$) or variability within groups.

Coaching evaluation criteria

Table 4 presents the results of an evaluation comparing three groups – LLM-power coaching (AI), human coaching (Human), and GPT Conversation (GPT) – on a set of variables related to coaching effectiveness. Pairwise

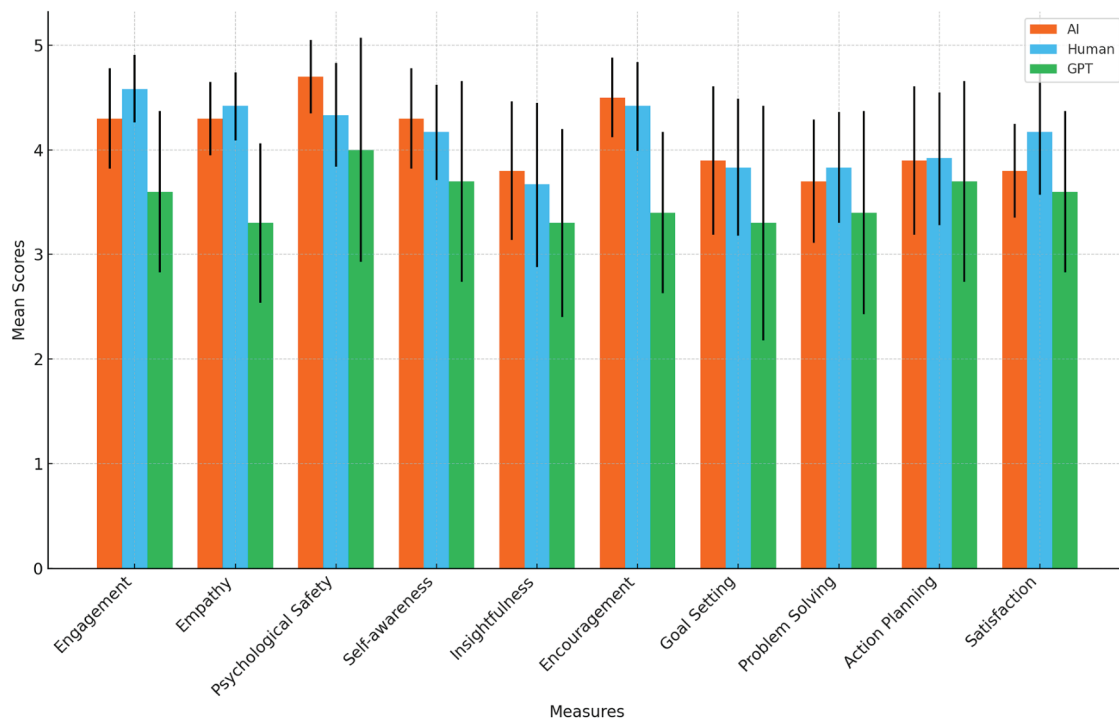


Figure 2. Comparison of measures across groups with 95% confidence intervals.

Table 4. Coaching evaluation criteria.

Measure	AI M (SD)	Human M (SD)	GPT M (SD)	AI v.s. GPT			AI v.s. Human			GPT v.s. Human		
				t	p	d	t	p	d	t	p	d
Engagement	4.30 (0.67)	4.58 (0.51)	3.60 (1.07)	1.74	0.1	0.82	-1.09	0.29	0.5	-2.65	0.02*	1.27
Empathy	4.30 (0.48)	4.42 (0.51)	3.30 (1.06)	2.72	0.02*	1.28	-0.55	0.59	0.24	-3.05	0.01*	1.46
Psychological Safety	4.70 (0.48)	4.33 (0.78)	4.00 (1.49)	1.41	0.19	0.67	1.35	0.19	0.58	-0.64	0.53	0.3
Self-awareness	4.30 (0.67)	4.17 (0.72)	3.70 (1.34)	1.27	0.23	0.6	0.45	0.66	0.2	-0.99	0.34	0.47
Insightfulness	3.80 (0.92)	3.67 (1.23)	3.30 (1.25)	1.02	0.32	0.48	0.29	0.77	0.13	-0.69	0.5	0.31
Encouragement	4.50 (0.53)	4.42 (0.67)	3.40 (1.07)	2.91	0.01*	1.37	0.33	0.75	0.14	-2.6	0.02*	1.22
Goal Setting	3.90 (0.99)	3.83 (1.03)	3.30 (1.57)	1.02	0.32	0.48	0.15	0.88	0.07	-0.92	0.37	0.43
Problem Solving	3.70 (0.82)	3.83 (0.83)	3.40 (1.35)	0.6	0.56	0.28	-0.38	0.71	0.17	-0.88	0.39	0.42
Action Planning	3.90 (0.99)	3.92 (1.0)	3.70 (1.34)	0.38	0.71	0.18	-0.04	0.97	0.02	-0.42	0.68	0.2
Satisfaction	3.80 (0.63)	4.17 (0.94)	3.60 (1.07)	0.51	0.62	0.24	-1.09	0.29	0.47	-1.3	0.21	0.59
Positive Affect	34.50 (7.92)	32.42 (8.70)	35.30 (11.90)	-0.18	0.86	0.08	0.59	0.56	0.26	0.64	0.53	0.3
Negative affect	14.80 (5.39)	13.50 (4.34)	17.10 (7.55)	-0.78	0.44	0.37	0.61	0.55	0.28	1.34	0.2	0.63

M=Mean; SD=Standard Deviation; t= t-test results for comparisons between two groups; p = p-value; *for p-value ≤ 0.05 ; d = effect size: d = 0.2 (small), d = 0.5 (medium), and d = 0.8 (large) (Cohen, 1988).

t-tests were conducted across various measures to compare differences between groups.

AI vs. GPT

Engagement. The AI group scored higher than the GPT group ($t = 1.74$, $p = 0.1$, $d = 0.82$), with a large effect size, though the difference is not statistically significant.

Empathy. A significant difference was observed, with the AI group scoring higher than the GPT group ($t = 2.72$, $p = 0.02 < 0.05$, $d = 1.28$), reflecting a large effect size.

Encouragement. The AI group significantly outperformed GPT ($t = 2.91$, $p = 0.01 < 0.05$, $d = 1.37$), with a large effect size.

Other measures (e.g. Psychological Safety, Self-awareness, Satisfaction): No significant differences ($p > 0.05$), though effect sizes for some measures (e.g. Psychological Safety) suggest medium trends.

These results suggest that AI-driven coaching consistently outperformed GPT conversation in measures like Empathy and Encouragement, with significant differences and large effect sizes. This highlights the greater efficacy of AI coaching compared to GPT in emotional and motivational domains.

AI vs. human

No significant differences were found across all measures ($p > 0.05$). The effect sizes for most comparisons were small to medium, with the highest observed in Engagement ($t = -1.09$, $p = 0.29$, $d = 0.5$), suggesting similar performance between AI and Human coaching.

These findings suggest that AI coaching achieves outcomes comparable to human coaching across all evaluated measures. This underscores the potential of AI as a viable alternative to traditional coaching.

GPT vs. human

Engagement. The Human group significantly outperformed GPT ($t = -2.65$, $p = 0.02 < 0.05$, $d = 1.27$), with a large effect size.

Empathy. The Human group scored significantly higher than GPT ($t = -3.05$, $p = 0.01 < 0.05$, $d = 1.46$), indicating a large effect size.

Encouragement. The Human group also significantly outperformed GPT ($t = -2.6$, $p = 0.02 < 0.05$, $d = 1.22$).

Other measures showed no significant differences, though effect sizes suggest medium trends for Satisfaction and Negative Affect.

Results indicate that Human coaching significantly outperformed GPT coaching in Engagement, Empathy, and Encouragement, with large effect sizes. This confirms the superiority of Human coaching over GPT for critical coaching dimensions.

Figure 2 provides a visualized comparison of mean scores among three groups across various coaching evaluation measures, with 95% confidence intervals indicated by error bars. Human coaching and AI coaching are generally more effective than GPT coaching in all measures, particularly in Engagement, Empathy, and Encouragement. Human coaching outperformed AI coaching in Engagement and Satisfaction, while AI coaching outperformed Human coaching in Psychological Safety though the differences appear less pronounced. For measures such as self-awareness, insightfulness, encouragement, goal setting, problem-solving, and action planning, AI coaching demonstrates comparable effectiveness to Human coaching. GPT showed relatively lower scores across most measures. GPT coaching consistently lags behind in most areas, with wider confidence intervals reflecting greater variability in its effectiveness. These results further reinforce significant differences in coaching

effectiveness among the groups, suggesting the potential for AI coaching as a viable alternative to traditional human coaching.

These results provide strong evidence supporting Hypothesis 1, as LLM-powered coaching outperforms ChatGPT-4 in key coaching dimensions. Additionally, the findings support Hypothesis 2, demonstrating that LLM-powered coaching achieves comparable effectiveness to traditional human coaching in most areas. These results highlight the potential of LLM-powered coaching as a scalable, effective alternative to human coaching for emotional and developmental support.

Qualitative analysis

This qualitative analysis focuses on participant feedback across the three groups: AI Coaching, Human Coaching, and GPT-4 Conversations. The data were analysed to identify key themes in participants' perceptions of the benefits, areas for improvement, and additional suggestions regarding their coaching experience.

Engagement and emotional depth

Human coaching was consistently praised for its ability to foster emotional connection and provide tailored guidance. AI coaching was recognized for encouraging self-reflection but was criticized for lacking emotional nuance. GPT-4, while convenient, struggled with emotional depth and often provided generic or repetitive responses.

Actionable outcomes

Both AI and human coaching excelled in helping participants create practical action plans, though AI participants suggested improvements in specificity and operational guidance. GPT-4 provided general suggestions but lacked the depth to drive meaningful behavioral change.

Accessibility and convenience

AI coaching and GPT-4 were noted for their accessibility, making them suitable for flexible, on-demand use. Human coaching, while impactful, was constrained by session time and scheduling.

The qualitative feedback highlights distinct strengths and areas for improvement for each coaching modality. Human coaching remains the gold standard for emotional depth and personalized support. AI coaching, while not as emotionally attuned as human coaching, shows promise as a structured, scalable alternative, particularly when personalized elements are enhanced. GPT-4 offers convenience and general guidance but

requires significant improvements in empathy and conversational depth to be considered a viable coaching solution.

Discussion

This study aimed to compare the effectiveness of LLM-powered coaching with traditional human coaching and ChatGPT-4 interactions across multiple dimensions of coaching effectiveness. The findings contribute to a growing body of literature examining the potential of artificial intelligence in coaching, offering valuable insights into the strengths and limitations of AI-driven coaching.

Effectiveness of LLM-Powered coaching

The results provide compelling evidence supporting Hypothesis 1: LLM-powered coaching outperforms ChatGPT-4 conversation. It consistently demonstrated significantly higher scores with large effect sizes in measures such as Empathy ($t = 2.72, p = 0.02, d = 1.28$), and Encouragement ($t = 2.91, p = 0.01, d = 1.37$). These findings are consistent with the qualitative feedback, where participants in the GPT-4 group frequently noted its lack of depth and emotional connection, with feedback highlighting repetitive responses and limited ability to engage in dynamic, context-sensitive interactions. In contrast, participants noted LLM-powered coach's structured responses and ability to foster actionable strategies and self-reflection. These findings suggest that prompt-tuned AI models incorporating psychological knowledge and coaching frameworks can bridge the gaps in generic AI capabilities, delivering coaching experiences that are more emotionally supportive and motivational.

Additionally, the results support Hypothesis 2, LLM-powered coaching performs comparably to human coaching. Across all measures, no statistically significant differences ($p > 0.05$) were found between LLM-powered coaching and human coaching. Notably, Hey Sherry even outperformed human coaching in Psychological Safety ($t = 1.35, p = 0.19, d = 0.58$), though this difference was not statistically significant. Psychological safety is a fundamental measure of a coach's ability to foster trust, allowing clients to take risks in learning and personal development (Ostrowski & Potter, 2023). According to the International Coaching Federation's (2019) core competencies, coaches should cultivate an environment that is safe, supportive, and conducive to open, uninhibited sharing. This may be attributed to the privacy inherent in the AI coaching setup, allowing participants to explore their thoughts, experiences, and feelings

without the fear of judgment or scrutiny from another person.

Qualitative data provided insights into the nuanced dynamics of interaction across coaching modalities. While LLM-powered coaching excelled in facilitating self-reflection and action planning, some participants described Hey Sherry's responses as formulaic and less emotionally nuanced. In contrast, human coaching received widespread praise for fostering emotional connection and relational depth. Emotional response, or affective resonance, refers to the ability to perceive and reflect on others' emotional states, often through signals such as emotional arousal (Decety, 2010; Decety & Cowell, 2014). While AI can analyze linguistic patterns to emulate human-like interactions, it lacks authentic emotional responses, limiting its ability to establish deep emotional bonds (Bachkirova & Kemp, 2024). Human coaches leverage nonverbal cues such as facial expressions, gestures, and vocal tone to enhance mutual understanding and acceptance, especially in conversations involving personal beliefs, values and aspirations (Bachkirova, 2016; Bachkirova & Kemp, 2024; De Haan & Gannon, 2017; Western, 2012). Leveraging emotional intelligence and depth, human coaches excel at navigating these complex and nuanced topics (Bachkirova & Kemp, 2024; Terblanche, Moly, De Haan, Nilsson, & Triberti, 2022). These findings suggest that while LLM-powered coaching is a highly effective tool for addressing specific, goal-oriented challenges, human coaching retains its irreplaceable role in cultivating meaningful emotional connections and facilitating relational dynamics enriched by nonverbal communication.

Emotional well-being outcomes

The analysis of Positive Affect and Negative Affect revealed meaningful trends in emotional well-being, though the results did not reach statistical significance. The LLM-powered coach demonstrated the greatest improvement in Positive Affect (+ 8.49%) with a small to medium effect size ($d = 0.33$) and the most substantial reduction in Negative Affect (− 27.45%) with a medium effect size ($d = -0.49$). In comparison, GPT-4 showed smaller emotional improvements (Positive Affect + 3.52%, $d = 0.12$; Negative Affect − 11.86%, $d = -0.36$), while human coaching had the smallest observed changes (Positive Affect + 1.83%, $d = 0.07$; Negative Affect − 8.47%, $d = -0.28$). These findings suggest that LLM-powered coaching may be particularly effective in promoting emotional well-being, outperforming both GPT and human coaching in measurable improvements.

Although these findings are promising, the lack of statistical significance still needs to be addressed. This

likely stems from three factors: (1) the small sample size ($N = 32$) reducing statistical power, (2) high variability in individual responses, evidenced by large standard deviations (e.g. Positive Affect post-coaching SD: AI = 7.92, Human = 8.70, GPT = 11.90), and (3) the short duration of the coaching intervention (20 minutes). Longer-term coaching may be required for measurable and sustained improvements in emotional well-being.

Advantages

LLM-powered coaching offers distinct advantages over traditional human coaching in terms of stable coaching maturity, accessibility, affordability, scalability, and language inclusivity. Hey Sherry delivers consistently high-quality coaching due to its rigorous design, which is grounded in a knowledge-based framework informed by prominent psychological theories, coaching models, and a specialized coaching dataset. In contrast, the effectiveness of human coaching heavily depends on the coach's competency and professional maturity (Terblanche, Moly, De Haan, Nilsson, & Triberti, 2022). Human coaches enter the field with varying levels of training and experience, ranging from minimal instruction to advanced post-graduate degrees, leading to diverse levels of coaching proficiency and maturity (Bachkirova & Smith, 2015; Bozer et al., 2014; Megginson & Clutterbuck, 2010).

Moreover, AI coaching operates 24/7, providing continuous access to coaching support without the constraints of scheduling or time-zone differences, which can be a barrier to traditional coaching. It is also a more affordable option, making professional-level coaching accessible to a broader audience. Furthermore, the scalability of LLM coaching enables it to support individual growth and well-being at scale, addressing the growing demand for personal development solutions while maintaining consistency in delivery and quality. Notably, LLM-powered coaching breaks language barriers by supporting over 80 languages, further extending its accessibility to a diverse, global audience and enhancing its inclusivity compared to human coaching. This combination of technological sophistication and broad accessibility positions LLM-powered coaching as a transformative tool for scalable coaching solutions.

Limitations

While the findings are promising, several limitations should be acknowledged. First, the study's small sample size ($N = 32$) reduced statistical power and generalizability. Second, the short duration of the coaching intervention (20-minute session) limited the impact of coaching

on emotional well-being and other measures. Additionally, the short-term outcomes raise uncertainty about the long-term effectiveness of LLM-powered coaching. Third, low survey completion rates underscore the need for enhanced strategies to improve participant engagement and retention in online interventions. Only 40.83% of the registered participants completed the pre-coaching surveys, and 26.67% successfully completed the entire study. In particular, the completion rate in the AI coaching group was lower than in the other two groups. To address this, additional participants were recruited for this group to balance the sample size across groups. Finally, the inclusion of only one human coach limits generalizability, as the effectiveness of human coaching is highly dependent on the coach's competency and professional maturity (Terblanche, Molyn, De Haan, Nilsson, & Triberti, 2022). Future studies should address the above limitations.

Future research

To maximize the potential of LLM-powered coaching systems, several key areas warrant further exploration. First, future research should include larger, more diverse participant groups to improve generalizability. Second, longitudinal studies are needed to assess the long-term effects of AI coaching on emotional well-being, behavior change, and coaching effectiveness.

Moreover, enhancing the understanding of emotional nuance, interactivity, and coaching depth is critical for advancing LLM-powered coaching systems. Future advancements should enable AI to interpret nonverbal cues such as tone, facial expressions, and gestures, fostering genuine emotional connections and coaching relationships. Affective computing technologies, including Speech Emotion Recognition (SER) and Multimodal Emotion Recognition (MER), offer promising pathways to improving AI's ability to navigate complex emotional dynamics and create more personalized, effective coaching experiences (Bilotti et al., 2024; Santoso et al., 2024).

Additionally, privacy and potential risks associated with AI coaching must also be addressed. A major concern is the opaque nature of LLM-generated outputs, often described as a 'black box', which creates uncertainty about reliability and potential risks (Demszky et al., 2023). Future studies should demystify these mechanisms, enhancing the transparency and trustworthiness of AI-driven coaching. Moreover, as LLMs process vast amounts of language data, safeguarding user privacy is critical. Researchers must implement stringent data security measures and privacy-preserving techniques to prevent unauthorized access or misuse of sensitive information (Demszky

et al., 2023). Another critical challenge is mitigating biases inherent in training data, which can lead to discriminatory or harmful outcomes, particularly for vulnerable populations (Demszky et al., 2023; Stade et al., 2024). Future research should focus on developing bias detection and correction mechanisms to ensure fairness in AI-driven coaching.

Finally, the lack of personhood and ethical reflexivity in AI systems presents a unique ethical dilemma. Unlike human coaches, AI lacks moral reasoning and accountability, making it unable to recognize or address risks associated with its input or advice (Bachkirova & Kemp, 2024). This absence of a duty of care, coupled with a lack of liability, raises concerns regarding responsibility for potential harm caused by AI coaching interactions (Mayhead, 2023; Wright & O'Connor, 2021). To address these issues, policymakers and researchers should establish clear frameworks that assign responsibility and accountability for AI-driven interventions, ensuring ethical considerations remain central to AI coaching innovations.

By addressing these areas – participant diversity, longitudinal impacts, emotional intelligence, privacy, bias mitigation, and ethical accountability – AI-powered coaching systems can become more effective, responsible, and transformative tools for personal and professional development.

Conclusion

This study provides compelling evidence that LLM-powered coaching is a highly effective alternative to both human coaching and GPT-4 interactions. Through an RCT, this research demonstrated that AI coaching significantly outperformed GPT-4 in key coaching dimensions such as empathy, engagement, and encouragement while achieving comparable effectiveness to human coaching across all measured criteria. Notably, LLM-powered coaching excelled in fostering Psychological Safety and led to the greatest improvements in emotional well-being, enhancing Positive Affect and reducing Negative Affect.

These findings highlight AI-driven coaching as an effective, accessible, and scalable solution that bridges gaps in traditional coaching availability, underscoring the viability of AI as a transformative tool in coaching. While AI coaching cannot fully replace human coaches due to its limitations in emotional nuance and nonverbal communication, it serves as a valuable complement, enhancing accessibility and expanding the coaching ecosystem, particularly for individuals with limited access to traditional coaching services.

Future research should expand participant diversity, investigate longitudinal impacts, and explore advanced technologies to refine AI coaching further. Addressing privacy, bias, and ethical concerns will be essential in ensuring the responsible and sustainable adoption of AI coaching.

Ethical statement

All participants were informed about the purpose of the study, the conditions of intervention, and the methods of data storage. Participation was voluntary. Ethical approval was obtained from the Institutional Review Board (IRB) at the University of Pennsylvania.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix A: Self-report coaching evaluation criteria

Coaching criteria	Questions
Engagement	(1) I felt engaged throughout the conversation.
Empathy	(2) The coach/GPT can understand my feelings and experience.
Psychological Safety	(3) I felt psychologically safe and comfortable sharing my thoughts, feelings, or experiences throughout the conversation.
Self-Awareness	(4) The conversation prompted me to explore my thoughts, emotions, or behaviors, and increased my self-awareness.
Insightfulness	(5) I discovered new insights or perspectives that I hadn't considered before the conversation.
Encouragement	(6) I felt encouraged and supported during the conversation.
Goal Setting	(7) The conversation helped clarify my personal or professional goals.
Problem Solving	(8) The conversation helped me generate effective solutions for addressing my current challenges or problems.
Action Plans	(9) The conversation assisted me in developing practical action plans.
Satisfaction	(10) Overall, I am satisfied with the conversation/coaching session.
Open-Ended Questions:	
	(11) What did you find most beneficial about the coaching session?
	(12) Were there any areas where you think the coaching could improve?
	(13) Please provide any additional comments or suggestions about your coaching experience.

Appendix B: Qualitative analysis of participants' perceived benefits, improvement areas, and suggestions across three coaching modalities

Group	Benefits Identified	Areas for Improvement	Additional Comments
A1	Comprehensive and Action-Oriented Feedback: Participants appreciated the clear, structured, and actionable strategies provided by the AI coach. Several noted that it helped them recognize what was happening internally and gave practical solutions. Empowering Self-Reflection: Many participants found value in the AI's questions, which facilitated deeper self-reflection and independent thinking. Statements like "It asks you to think for yourself" and "It really helped me put my thoughts together" reflect this benefit. Flexibility and Accessibility: Participants highlighted the flexibility of the AI coach in being available at their convenience and offering diverse advice for various scenarios.	Personalization and Depth: Participants mentioned that the responses felt "formulaic" or "not personalized enough," with repeated phrases or generic advice. Suggestions for more tailored and emotionally nuanced responses were common. Interactivity: Feedback indicated a desire for deeper engagement and interaction. Participants noted the need for the AI to "ask fewer questions in every statement" and provide more dialogic, conversational feedback rather than a rigid format. Practical Action Plans: While the AI provided general actionable strategies, some participants suggested the need for clearer timelines and guidance to better operationalize action plans.	Overall, participants appreciated the AI coach as a helpful tool for relieving stress and providing clarity in problem-solving. However, they suggested that adding life-like and interactive elements could significantly enhance the experience
Human	Emotional Connection and Depth: Participants valued the coach's ability to ask thought-provoking questions, affirm their feelings, and provide personalized attention. For example, one participant highlighted, "The coach asked questions about the underlying reasons for my behavior." Support for Self-Awareness and Goal Clarity: Many participants emphasized that human coaching helped them clarify their thoughts, create actionable plans, and reflect on their inner challenges. Statements like "It challenged me to identify a path to achieving my new career goals" exemplify this benefit. Positive Emotional Impact: The coach's empathy and validation fostered a sense of comfort and inspiration among participants, as reflected in comments such as "It made me feel at ease and more comfortable."	Session Duration: Several participants noted that the sessions felt too short, limiting the depth of exploration. Feedback such as "If we had more time, it would have been more beneficial" was common. Resource Generation: Some participants suggested that more follow-up materials or resources could further enhance the impact of coaching.	Participants expressed gratitude for the human coaching experience, emphasizing its positive emotional tone and practicality in achieving actionable outcomes.

(Continued)

(Continued).

Group	Benefits Identified	Areas for Improvement	Additional Comments
GPT	Convenience and Availability: Participants appreciated GPT-4's accessibility, noting that it was "available anytime and anywhere" and allowed for open, unfiltered conversations. Support for Independent Thinking: Some participants found GPT-4 helpful in organizing their thoughts and reducing anxiety. Comments such as "It let me clarify what I was thinking" highlight this strength. General Insights and Suggestions: Participants valued the practical insights and structured responses GPT-4 provided.	Empathy and Emotional Depth: Many participants pointed out that GPT-4 lacked the emotional understanding and empathy required to engage deeply. Comments like "It lacks the empathy and emotions required for a human" were frequent. Engagement and Interactivity: Participants found the interaction "not personal" and "too repetitive." Some noted the need for more dynamic responses that could drive the conversation forward rather than just reacting. Visual and Contextual Elements: Suggestions for incorporating visual aids or more personalized, contextually aware responses were common.	While participants appreciated GPT-4's practicality, many felt it could not replicate the depth and emotional connection provided by human coaches. Some noted that the experience was "not the best," but still valuable for certain scenarios.